

ECE 447

Lesson 22

Digital

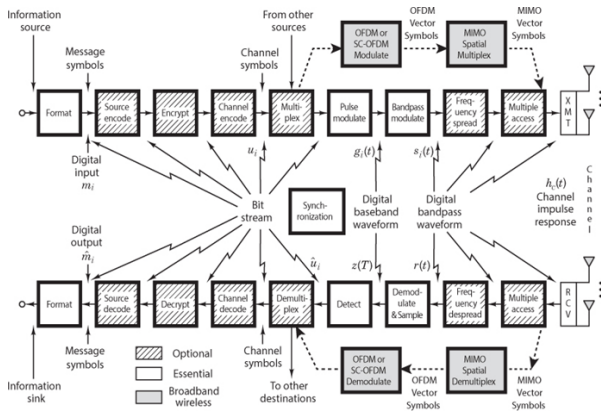
Communication and Line Coding



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW4.** Was due Lesson 21 (6 Oct).
 - **HW5.** Due Lesson 23 (Tue, 13 Oct) to Gradescope.
 - **Lab 5.** Due Lesson 24 (Thu, 15 Oct) to Gradescope.

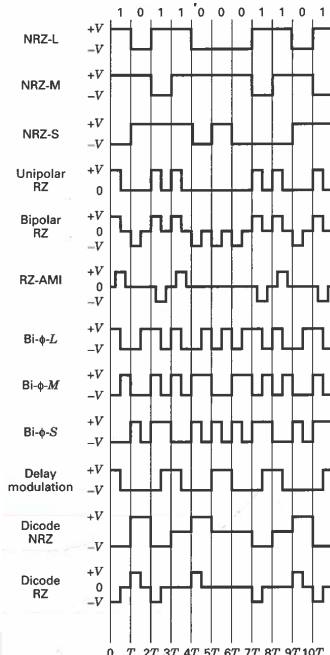
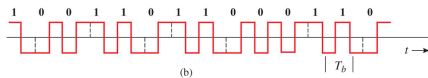
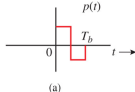
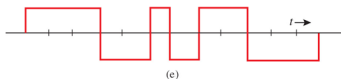
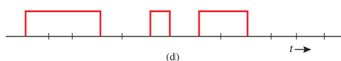
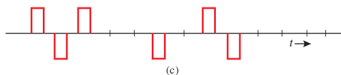
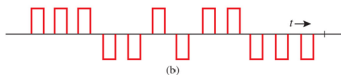
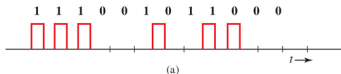


BASEBAND LINE CODING

Desired line code properties:

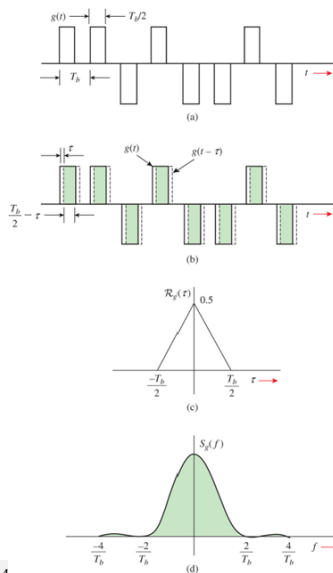
- Low bandwidth
- Power efficiency - low tx power for given bandwidth and error detection rate
- Error detection/correction capability
- No DC content
- Timing info (self-clocking)
- Transparency

TYPES OF LINE CODES



LINE CODE PSDs

Example 3.24 - Our good friend, Autocorrelation!



LINE CODE SPECTRA

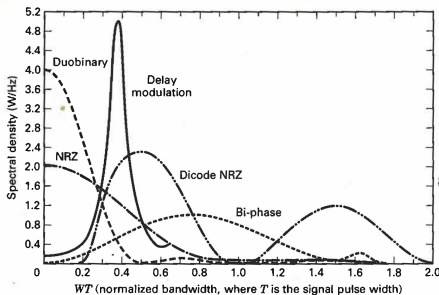
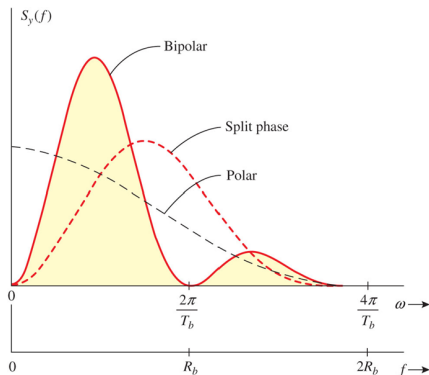


Figure 2.23 Spectral densities of various PCM waveforms.

EXAMPLE: CHANGE THE PULSE, KEEP THE CODE

- The pulse shape and the line code enter the PSD **separately**: $S_y(f) = \frac{|P(f)|^2}{T_b} \times (\text{code factor})$
 - Polar: 1 On-off: $\frac{1}{4} \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta \left(f - \frac{n}{T_b} \right) \right]$ Bipolar: $\sin^2 (\pi f T_b)$
 - Change the pulse and **only** $|P(f)|^2$ moves — the code factor stays put
- Try it.** Polar signaling with a *half-width* RZ pulse, $p(t) = \Pi \left(\frac{2t}{T_b} \right)$. You need $\Pi(t/\tau) \leftrightarrow \tau \text{sinc}(\pi f \tau)$.
 - What is τ here? _____ So $P(f) =$ _____
 - $|P(f)|^2 =$ _____ $S_y(f) =$ _____
 - First null where $\pi f T_b / 2 = \pi$, so $f =$ _____
- Full-width NRZ, $\Pi(t/T_b)$, nulls at R_b . Halving the pulse width _____ the bandwidth.
- HW6 Problem 1 runs this same recipe on a different pulse: find the pair, square it, multiply.

LINE CODE TO BASEBAND WAVEFORM

